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Beyond Anthropomorphism: Social Presence in Human–AI Collaboration Processes

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ABSTRACT Artificial intelligence (AI) systems, evolving from reactive tools to proactive collaborators, reshape team dynamics in today's digital workplaces. Text-based collaboration now frequently involves AI participants that perform tasks traditionally handled by humans, such as creative problem-solving and decision-making. This transition has been linked to changes in group dynamics, particularly in relation to social presence, which appears to shape the patterns of productivity and collaboration. We conducted three empirical studies on human–AI teams to investigate the relationship between social presence and willingness to depend on teammates, team-oriented commitment, and motivation to contribute. Drawing on social presence theory and theory of planned behaviour, our results show that while social presence has a direct association with motivation to contribute, an equally important indirect pathway is associated with human factors like team-oriented commitment and team members' willingness to depend on each other. We show that while social presence is significantly associated with behavioural intentions, greater AI familiarity and understandability are associated with a stronger relationship, raising questions about the sufficiency of relying solely on anthropomorphic features. Our study contributes to the understanding of human–AI collaboration in social presence research, highlighting the importance of considering social and interpersonal processes in hybrid teams. Our findings have managerial implications for organizations looking to adopt AI-based systems for collaboration.

Keywords: artificial intelligence, augmentation, future of work, human–AI collaboration, social presence theory, theory of planned behaviour

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INTRODUCTION

The rapid rise of artificial intelligence (AI) is transforming workplaces, with organizations increasingly embedding it into daily operations (Anthony et al., 2023; Hevner and Storey, 2022; Jia et al., 2023). As AI transitions from a reactive tool to a proactive collaborator (Baird and Maruping, 2021; Diebel et al., 2025), it is reshaping the nature of team interactions (Dennis et al., 2023; Diebel et al., 2025). For instance, text-based AI assistants, powered by large language models (LLMs) like ChatGPT and Claude, are now seamlessly integrated into platforms such as Slack[1] and Microsoft Teams,[2] enabling routinized AI collaboration. They reflect an evolving trend towards text-based interfaces to facilitate human–AI collaboration,[3] often accompanied by shifts in team structure and dynamics as new agentic actors are embedded into established workflows (Faulconbridge et al., 2025). Such systems seek to replace or augment human labour beyond traditional information systems (IS) that simply support human activities (Berente et al., 2021; Rahwan et al., 2019). For example, LLMs can now autonomously clarify information, guide users, and manage collaborative scenarios, reflecting the shift from reactive to proactive behaviour (Deng et al., 2023; Diebel et al., 2025). Additionally, these systems can actively critique inputs and refine their outputs through processes resembling human teamwork, where decisions are refined through collaborative deliberation and debate (Gordon, 2023).

This evolution is increasingly evident in real-world applications, like Microsoft's Copilot, which is regularly updated with new features to support collaboration, including analysing meeting transcripts and offering proactive suggestions.[4] Furthermore, AI is rapidly taking on functions typically associated with humans, including creativity, learning, and reasoning (Cropley et al., 2021; Jia et al., 2023; Shaikh and Cruz, 2023). Beyond these advanced capabilities, AI is increasingly contributing to value creation tasks like collaborative ideation (Aleksander, 2017; Jia et al., 2023; Senoner et al., 2022; Siemon, 2022). This positions AI as a teammate rather than a passive tool – an intelligent agent that actively engages with human teams, delivering dynamic, context-aware contributions that derive meaning from their embeddedness in situated interaction rather than from mere cognitive emulation (Anthony et al., 2023; Pakarinen and Huising, 2025; Seeber et al., 2018, 2020; Siemon and Strohmann, 2021). These contributions increasingly depend not only on technical capabilities but also on how AI systems are integrated into the fabric of collaborative practices and interpreted within social contexts (Pakarinen and Huising, 2025). This transformation not only alters task execution but also raises important questions about the evolving dynamics of collaboration and professional work. While much of the discourse has focused on macro-level consequences, there is increasing recognition of the need to understand how organizations and individuals interact with AI systems and reshape collaboration norms in practice (Sarala et al., 2025). Understanding the evolving nature of human–AI collaboration requires attention to the micro-level mechanisms through which it reshapes team dynamics and may enhance productivity in digital workplaces.

Nevertheless, the idea of AI assuming human roles as collaborators and teammates comes with its own distinct challenges, such as job displacement and disruption of

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established human activities (Anderson et al., 2018; Schwartz et al., 2019). The presence of AI teammates is associated with attitudinal and behavioural changes like diffusion of accountability, where participants cede responsibility to their AI teammates (Siemon and Wank, 2021; Stieglitz et al., 2021). These challenges illustrate associations observed with the integration of AI into teams, not only on individual roles but also on the collective principles that underpin effective collaboration. Therefore, incorporating AI as teammates necessitates a thoughtful re-evaluation of established norms and dynamics that have historically governed human teams, particularly in how collaboration may affect collective outcomes (Bansal et al., 2019; Seeber et al., 2019, 2020; Siemon and Strohmann, 2021).

The fundamental purpose of collaboration is to enhance mutual learning and augment the value of shared outcomes (Bunderson and Sutcliffe, 2002; Chae et al., 2015). To achieve this, teams must cultivate an environment that fosters interactions conducive to collaboration (de Vreede and Briggs, 2019; Randrup et al., 2018). The absence of positive interactions between team members may impede effective collaboration regardless of the competencies of the individual team members (Fabriek et al., 2008; Mohammed and Angell, 2003). In human–AI collaboration, these dynamics are further complicated by the inherent differences between human and AI teammates, as AI lacks the emotional and contextual cues of human collaborators. While there are fairly established norms for relationships between human team members, the expectations for human–AI relationships are less well defined, which may create dissonance among human team members (Shaikh and Cruz, 2023; Zhang et al., 2021). Collaboration in such settings cannot be reduced to predefined roles or static competencies but must be understood as emerging from relational and situated dynamics shaped through interaction (Pakarinen and Huising, 2025). Therefore, it is critical to identify the dynamics that guide these collaborations to design and facilitate productive human–AI collaboration.

Social presence, that is, an individual's perception of the presence of another entity, plays a crucial role in digital collaboration dynamics (Lim, 2018). Perceived social presence of AI systems is significantly correlated with enhanced system usage and overall acceptance (Elshan et al., 2022), highlighting the importance of perceptual salience in digital collaborations (Gupta et al., 2023; Gupta and Bostrom, 2006). Unlike human teammates who can draw on a wide range of verbal and non-verbal cues to establish their presence, AI teammates interact exclusively through digital channels, resulting in fundamentally different dynamics of social presence. Evaluating how AI is associated with perceptions of social presence in text-based collaboration platforms is critical as it is directly linked to team trust, engagement, and motivation (Munnukka et al., 2022; Ruiz-Equihua et al., 2023). Given the absence of physical or emotional cues, understanding these dynamics is vital for designing AI systems that aim to seamlessly integrate into human teams and improve collaborative outcomes.

To this end, this paper addresses the following research question: *How is the social presence of an AI teammate related to team members' motivation to contribute to team efforts in human–AI teams?* We build on social presence theory (SPT) and theory of planned behaviour (TPB) to investigate the complex role of AI social presence in collaborative settings involving text-based brainstorming. Specifically, we challenge the assumption of a direct relationship between social presence and motivational contributions within teams, particularly

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in human–AI collaborations, and argue for a more nuanced association. To this end, we conducted a series of three studies to empirically examine the relationship between social presence and team members' motivation to contribute and to unpack the underlying mechanisms of this relationship.

Our research contributes to the literature on human–AI collaboration by offering insight into how social presence, a well-studied construct in human–human collaboration, operates within the unique context of human–AI teams. While prior research has explored social presence in digital environments, the application of social presence theory to human–AI collaboration remains underexplored, particularly in terms of how AI's social presence is associated with team motivation and productivity. Given the rapid organizational adoption of human–AI collaborative systems with significant implications for team effectiveness, understanding how AI's social presence is linked to team members' motivation to contribute is essential for both theoretical advancement and practical implementation. Accordingly, this research extends the understanding of social presence by contextualizing it within the evolving dynamics of human–AI collaboration. It explores the ways in which AI can effectively integrate into human teams without being associated with lower levels of motivation or productivity. Our findings further contribute to the literature on creative collaboration by identifying associations between AI teammates and key team processes such as trust, commitment, and willingness to contribute. In doing so, we contribute to broader perspectives that recognize AI integration as a process that reshapes collaboration by introducing new forms of agency and shifting the conditions under which professional and team-based work unfolds (Faulconbridge et al., 2025). Finally, our research also offers practical implications for the design of AI systems that foster trust, engagement, and sustained productivity in human–AI teams. In the remainder of the paper, we first outline our theoretical foundations and develop our hypotheses. Next, we provide details on each of our studies. We conclude the paper by discussing the implications of our findings, limitations, and directions for future research.

THEORETICAL FOUNDATIONS

Creative collaboration – joint effort towards generating novel ideas and artefacts – is a critical aspect of modern teamwork (Kolfschoten et al., 2012; Randrup et al., 2018). Brainstorming is one such collaboration technique. While it can be done verbally in in-person settings, text-based collaboration tools such as Microsoft Teams or Slack are becoming equally popular for brainstorming (Kamonmarttayakul et al., 2021; Montrief et al., 2020). Research on electronic brainstorming in the 1990s (Connolly et al., 1993; Fjermestad and Hiltz, 1998; Gallupe et al., 1992; Nunamaker et al., 1987) shows that text-based and electronic brainstorming methods can significantly reduce production blocking and other negative group dynamics compared to verbal interaction, thereby improving the effectiveness of idea generation and overall creative collaboration. Productive collaboration occurs when all team members consciously aim to achieve the team's goals and act purposefully towards those goals (Grant and Shandell, 2022). As a result, the quality of online collaboration processes and

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outcomes relies on the behavioural intention of individual team members (Glover and Kim, 2021; Randrup et al., 2018).

Theory of Planned Behaviour (TPB)

Theory of planned behaviour (TPB) represents an explanatory and predictive theory of attitudinal-behavioural reactions based on three factors that predict an individual's intention to engage in a behaviour at a specific time and place: personal attitudes, subjective norms, and perceived behavioural control (Ajzen, 1991). TPB explains how perceived behavioural control can influence actions in situations where one may not have complete volitional control over their behaviour. In other words, when performing a specific action, an individual's actual behaviour is directly influenced by their behavioural intention, which, in turn, is determined by attitudes and norms. Thus, behavioural intention indicates a person's willingness to perform a specific behaviour (Ajzen, 1991). TPB has been used in many IS and management studies to explain user adoption behaviour (e.g., Bhattacherjee, 2000; Mathieson, 1991; Pedersen, 2005; Taylor and Todd, 1995; Yang et al., 2016). In addition, the influence of TPB is evident in the technology acceptance model (TAM), the unified theory of acceptance and use of technology (UTAUT), and several other extended models to predict behavioural intentions (e.g., Aboelmaged, 2010; Curras-Perez et al., 2014; Zhang and Aikman, 2007).

In collaborative settings, TPB helps to explain and predict how team members' attitude – specifically their motivation to contribute – is influenced by the surrounding social environment and group dynamics. During collaborative ideation, behavioural intention is often linked to an individual's motivation to contribute. For instance, TPB can explain how an individual's intentions influence task completion in creative scenarios (Al-Lozi and Papazafeiropoulou, 2012; Wang et al., 2015; Zielińska-Tomczak et al., 2021). In this context, individual attitudes and subjective norms, both of which are influenced by the presence of team members, are directly associated with shaping behavioural intentions (Haines, 2021; Lount et al., 2008). These factors are particularly relevant in human–AI collaborations where AI systems are increasingly influencing team norms and dynamics.

Therefore, we propose TPB as an explanatory lens to understand the influence of team members' behavioural intentions during human–AI collaborations. We use TPB as an overall framework to understand the association of attitudinal and normative factors with the motivation to contribute to collaborative ideation in human–AI teams.

Social Presence Theory (SPT)

The term 'social presence' was coined by Short et al. (1976) as 'the degree of salience of the other person in the communication and the consequent salience of the interpersonal relationships' (p. 65). Social presence, conveyed by characteristics like human contact, person ability, human warmth, sociability, and human sensitivity, is a fundamental human need (Short et al., 1976). In this study, we operationalize social presence as the extent to which an actor is perceived as a 'real person' within

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collaborative computer-mediated communication environments (Gunawardena and Zittle, 1997).

SPT offers an explanatory foundation for the influence of team members' behavioural intentions in collaboration scenarios with AI teammates. It describes the manner in which entities exist in the digital world and react to users on digital platforms (Heater, 1992). According to SPT, when two people interact, they engage in specific role-based activities while developing or maintaining personal connections via interpersonal exchanges. The social presence of the users exerts a considerable influence on other users' behaviour and attitudes in a group setting (Short et al., 1976). Even when users are connected via technology, a high degree of social presence is likely to enhance interaction and collaboration among team members (Wei et al., 2017) by influencing participant engagement and the motivation to perform within teams (Molinillo et al., 2018). In fact, a strong social presence of human teammates can increase the behavioural intention to contribute (George, 1992; Haines, 2021). However, social presence does not occur in isolation and is influenced by factors such as psychological involvement and behavioural commitment (Biocca et al., 2003). Psychological involvement occurs when an observer becomes aware of another person's intention or idea by focusing on the other person's emotional state. Behavioural commitment occurs when an observer notices a change in another person's perceived reliance, connection, and responsiveness to the observer's activities (Qiu and Benbasat, 2009). Thus, in traditional human–human collaboration, a high degree of social presence typically leads to enhanced interaction, greater commitment, and increased motivation to contribute (Molinillo et al., 2018; Srivastava and Chandra, 2018).

However, in human–AI collaborations, the challenge lies in how AI systems, which lack physical embodiment and emotional nuance, can simulate social presence in meaningful ways. AI systems, particularly those integrated into platforms like Slack and Microsoft Teams, interact with human teammates through text-based communication, mimicking aspects of human interaction (Diederich et al., 2022; Elshan et al., 2022). While these AI systems attempt to foster a sense of presence by interpreting context and providing proactive responses (Qiu and Benbasat, 2009; Seeger et al., 2021), they still trigger a lower social presence compared to humans (Araujo, 2018; Feine et al., 2019; Nass and Moon, 2000a), as they lack critical human characteristics. This is particularly evident in text-based collaboration scenarios where other communication modalities are not present (i.e., voice or body language) (Seeger et al., 2021). Lower social presence of conversational agents is associated with reductions in quality interactions and satisfaction (Verhagen et al., 2014), perceived enjoyment (Qiu and Benbasat, 2009), trust, personal attachment (Lee et al., 2021), and emotional connection (Araujo, 2018).

Human–AI Collaboration Through the Lens of SPT and TPB

Existing theories in collaboration and motivation, such as SPT and TPB, are typically centred on human–human collaboration. However, the integration of AI into teams results in more complex interactions that differ from conventional expectations (Dennis et al., 2023). In recent years, studies have explored the dynamics of

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human–AI collaboration, focusing on how AI systems transition from tools to teammates in collaborative work (Dennis et al., 2023; Hevner and Storey, 2022). AI teammates, in contrast to assistant chatbots or other forms of automation, engage in complex decision-making processes and contribute autonomously to team tasks (Siemon, 2022). Empirical research at the intersection of AI and team dynamics confirms the opportunities and challenges of integrating AI into teams (Dennis et al., 2023). For example, AI can enhance decision-making efficiency, but it also often is accompanied by reduced trust and cohesion among human team members, thereby diminishing motivation in collaborative tasks (Baird and Maruping, 2021; Glikson and Woolley, 2020; Seeber et al., 2020). Building on these studies, our research contributes to the scientific discourse by specifically examining the proactive role of AI teammates in team settings, focusing on how their autonomous behaviour influences social presence and team dynamics.

We draw on SPT and Baird and Maruping's (2021) contention that IS artefacts are evolving from passive tools to proactive entities capable of initiating action and assuming responsibilities. This evolution necessitates a re-evaluation of the traditional user-IS artefact dynamic, particularly concerning how social presence – the sense of being with another – is cultivated and perceived in AI interactions. Traditionally, social presence has been understood as the degree to which a communicator is perceived as a real person in mediated communication (Heater, 1992; Short et al., 1976). This conceptualization relies on human-like attributes such as warmth, personalization, and immediacy, which are naturally present in human interactions but are difficult to fully emulate in AI-driven communications. While today's AI capabilities are sophisticated, they often struggle to replicate the full range of human social cues that facilitate a genuine sense of social presence (Chaturvedi et al., 2023). This limitation means that AI interactions lack the depth and authenticity of human interactions, and not all AI features may result in enhancing their social presence (Seeger et al., 2021). That is, while AI may simulate the aspects of social presence, its impact on human team members' motivation may not be as powerful as human–human interaction.

The anthropomorphic design of AI agents can significantly enhance perceptions of social presence by embedding human-like linguistic cues and emotional responsiveness into AI interactions (Konya-Baumbach et al., 2023; Seeger et al., 2021). Rather than simply mimicking human interaction, this design strategically leverages human-like characteristics to create a new dimension of perception and experience, where AI agents are perceived not just as tools but as social actors capable of engaging in meaningful interactions (Chaturvedi et al., 2023; Seeger et al., 2021). This shift is also supported by theories like the theory of mind and social response theory, which suggest that users attribute mental states to AI agents when they exhibit behaviours indicating understanding and responsiveness (Nass and Moon, 2000b; Premack and Woodruff, 1978). By embedding these traits, AI agents can foster a sense of intentionality and emotional engagement, thereby enhancing their role as social actors in interactions (Epley et al., 2007; Konya-Baumbach et al., 2023).

Similarly, TPB offers a broader lens in which behaviour is a product of attitudes, subjective norms, and perceived behavioural control (Ajzen, 1991). Specifically, team members' motivation to collaborate is influenced by their beliefs about the outcomes

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of collaboration, the expectations of other team members, and their perceived control over the collaborative task (Mazzetti and Schaufeli, 2022). The TPB framework is useful to understand how the AI presence may elicit different interpretative and evaluative processes by team members. For instance, when working with AI teammates, the novelty and unpredictability of AI behaviour could affect team members' perceived behavioural control, potentially diluting the impact of AI's perceived social presence on motivation (Baek and Kim, 2023).

HYPOTHESES DEVELOPMENT

To study social presence in human–AI collaboration, we developed a conceptual model consisting of four constructs: social presence, willingness to depend on others, team-oriented commitment, and motivation to contribute. This section provides an explanation of the corresponding hypotheses. The overall model is depicted in Figure 1.

Human and AI Social Presence

Socioemotional features are crucial design considerations for collaborative technologies (Molinillo et al., 2018). Social presence is one such feature that has major implications for collaboration artefact design (Molinillo et al., 2018; Srivastava and Chandra, 2018). Social presence is the degree to which an entity is perceived as a 'real' social actor. It is an individualized experiential phenomenon, with different collaborators experiencing varying amounts of social presence when interacting within the same collaboration system (Wei et al., 2017). At the highest end of the continuum, social presence is synonymous with another person being in the room with the team member. At the lowest end, there is a complete sense of isolation from the team. The degree of social presence often depends on factors like choice of the technology medium (Rice, 1993), familiarity with teammates (Hudson and Cairns, 2016), and the level of interactions (Lee and Huang, 2018). Consequently, social presence on a digital medium is likely most palpable when team members are familiar with each other and communicate with each other consistently.

When new or unfamiliar team members are introduced in less visual digital settings, such as text-based collaborations, it becomes challenging for existing members to understand

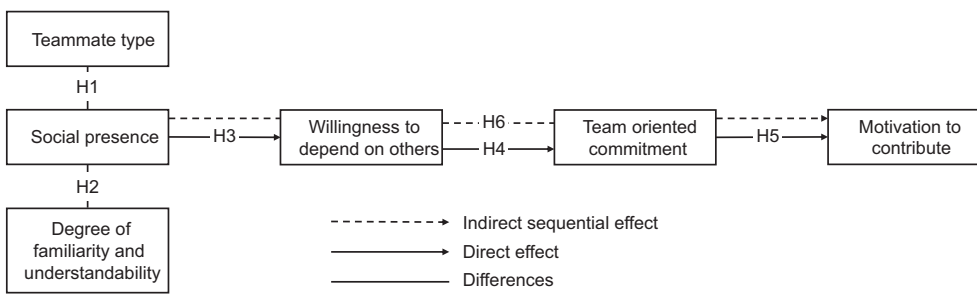

Figure 1. Conceptual model

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new teammates' intentions or even to be aware of their presence (Gruenfeld et al., 1996; Harrison et al., 2003). This difficulty is particularly pronounced with AI actors, given their lack of familiarity to users compared to human actors (Horowitz et al., 2024). Consequently, team members may perceive AI actors as less personal, sociable, and sensitive, making them reluctant to engage with AI teammates (Horowitz et al., 2024; Komiak and Benbasat, 2006; Verhagen et al., 2014). Accordingly, we hypothesize the following:

> *H1:* Team members feel lower social presence from AI teammates than they do from human teammates.

AI Familiarity and Social Presence

Anthropomorphic designs allow AI-based systems to have human characteristics (particularly social properties) (Dubosc et al., 2021). As highlighted in previous studies, social presence is often enhanced through the incorporation of anthropomorphic design features (Dubosc et al., 2021; Elshan et al., 2022). Notably, users report higher social presence when they engage in a human-like interaction with AI entities (Lee et al., 2021). Consequently, AI-based systems that exhibit human behaviour, like communicating in natural language, are likely to create a greater social presence (Diederich et al., 2022; Elshan et al., 2022; Lee et al., 2021; Qiu and Benbasat, 2009). However, the efficacy of these features is constrained to the quality of text-based collaboration platforms due to the absence of human-like cues such as virtual embodiment (including body language and facial expression) or voice-based communication, which play critical roles in increasing social presence (Feine et al., 2019). Consequently, while anthropomorphic designs can enhance social presence, the limitations inherent in text-only environments may prevent these systems from reaching their full potential in fostering deep user engagement and effective collaboration.

However, an understanding or familiarity of how AI systems operate is also likely to increase the AI's social presence (Biocca et al., 2003; Lee et al., 2021; Tu and McIsaac, 2002). As familiarity develops through repeated and situated interactions, it allows users to form expectations and interpret AI behaviour in context, reinforcing the sense of presence in collaborative settings (Pakarinen and Huising, 2025). Familiarity is the understanding about things and often refers to previous direct and indirect interactions, experiences, and knowledge (Gefen, 2000; Komiak and Benbasat, 2006). Familiarity with AI-based systems has a significant positive impact on trust (Mahmud et al., 2022), as it reduces the uncertainties associated with expectations of and use of such systems through a better understanding of what has happened in the past (Gefen, 2000; Komiak and Benbasat, 2006). An understanding of how AI-based systems operate and the logic they follow allows team members to better interact with them. When team members are familiar with an AI-based system and understand its functioning and behaviour (Lee et al., 2021; Van Hooijdonk, 2021), their perception of its social presence may increase. People tend to feel a higher social presence towards familiar and understandable AI team members compared to unknown and unfamiliar ones, as the former are perceived as more relatable and predictable (Lee and Nass, 2003; Nass and Moon, 2000a). In contrast, unknown and unfamiliar AI team members may

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evoke feelings of apprehension and uncertainty, as their decision-making processes and behaviour can appear less transparent and more difficult to comprehend (Turkle, 2011). Accordingly, we hypothesize:

> *H2:* Team members feel higher social presence toward familiar and understandable AI teammates than they do toward unknown and unfamiliar AI teammates.

AI Social Presence and Willingness to Depend on Others

SPT has been widely recognized as a framework for understanding the dynamics of online collaboration and learning environments (Garrison et al., 2001). When team members experience a high level of social presence, they are more likely to feel connected, engaged, and committed to the collaborative process (Sung and Mayer, 2012; Wang et al., 2022). Extending this concept to human–AI collaboration, we explore how the social presence attributed to AI systems influences team dynamics, distinct from the general social presence of the team. This differentiation is crucial as AI systems, despite not being human, can exhibit traits that may be perceived as personal and sociable, impacting collaborative processes similar to human team members (Kumar and Benbasat, 2002; Lee and Nass, 2003). In the context of AI teammates, social presence can foster a sense of trust and psychological safety (Catyanadika and Rajasekera, 2022). When individuals perceive AI as a supportive and reliable teammate, they are more likely to feel comfortable sharing ideas, taking risks, and relying on the contributions of both human and AI teammates (Bao et al., 2021).

Moreover, the social presence of AI can facilitate the development of interpersonal relationships and emotional bonds within virtual teams (Kreijns et al., 2003). When team members experience a strong sense of social presence, they are more likely to engage in social interactions, share personal experiences, and develop a sense of camaraderie with both human and AI teammates (Haines, 2021; Rourke et al., 1999). These emotional connections can enhance team cohesion and increase individuals' willingness to depend on their teammates, including AI, during online collaborative ideation activities (Garrison et al., 2001; Molinillo et al., 2018). As team members feel more connected and invested in one another, they are more likely to trust and rely on each other's ideas and contributions, regardless of whether they come from human or AI teammates (Kreijns et al., 2003).

TPB also provides valuable insights into the relationship between the social presence of AI and willingness to depend on teammates in online collaborative ideation (Ajzen, 1991). Social presence can positively influence team members' attitudes towards depending on both human and AI teammates, as they perceive AI as a real and trustworthy collaborator (Fui-Hoon Nah et al., 2023). Furthermore, the social presence of AI can contribute to the development of subjective norms that support interdependence and collaboration within the team, including reliance on AI (Molina and Sundar, 2022). Finally, social presence can promote a sense of accountability and responsibility among team members during an online collaborative ideation (Zhu et al., 2010). When individuals perceive AI as a real and present teammate, they are more likely to feel accountable for their own contributions and responsible for supporting the team's collective efforts, which include AI's contributions (Seeber et al., 2020). This sense of accountability can increase team members' willingness to

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depend on one another, including AI, as they recognize the importance of their individual roles in the collaborative ideation process (Maaravi et al., 2021).

Finally, social presence can facilitate effective communication and knowledge sharing in text-based online collaborative ideation (Garrison et al., 2001; Sharma et al., 2023; Wang et al., 2022). When team members experience a high level of communication from the AI, they are more likely to engage in active and meaningful discourse, ask questions, and provide feedback to both human and AI teammates (Seeber et al., 2020; Zhang et al., 2023). This open and interactive communication can foster a shared understanding among team members and enhance their willingness to depend on one another's expertise and insights, including those provided by AI (La Torre et al., 2021; Piorkowski et al., 2021). Consequently, we hypothesize that social presence is positively associated with the willingness to depend on others.

> *H3:* Team members' social presence of AI is positively associated with the willingness to depend on teammates in online collaborative ideation activities.

Willingness to Depend and Commitment

The success of collaborative ideation depends on various factors, including team members' commitment to the process (Nijstad and De Dreu, 2002). One dynamic that can significantly influence team-oriented commitment to collaborative ideation is the willingness to depend on others (Bishop et al., 2000; Bishop and Scott, 2000). Willingness to depend on others is closely linked to trust and psychological safety within a team (Edmondson, 1999). When individuals are willing to rely on their teammates, they feel more comfortable sharing ideas, taking risks, and engaging in collaborative ideation activities (Edmondson et al., 2004). This trust fosters a sense of commitment to the team and its goals, as members feel supported and valued (Bishop et al., 2000). Moreover, when individuals demonstrate a willingness to depend on others, it can create a sense of reciprocal commitment within the team (Bishop and Scott, 2000). As team members observe their colleagues' willingness to rely on and support one another, they are more likely to feel a strong sense of commitment to the team and its collaborative ideation efforts (Pearce and Gregersen, 1991).

Dependence on others also implies a sense of shared responsibility for the team's outcomes (Pearce and Gregersen, 1991). When individuals are willing to rely on their teammates, they recognize that the success of collaborative ideation activities depends on the collective efforts of the team (Van der Vegt and Janssen, 2003). This shared responsibility can increase team-oriented commitment, as members feel accountable for contributing to the team's goals (Bishop et al., 2000). Furthermore, interdependence and cohesion within the team are strengthened when individuals are willing to depend on others (Van der Vegt and Janssen, 2003). As team members recognize the importance of relying on their teammates' skills and contributions, they are more likely to develop a strong sense of unity and commitment to the team's collaborative ideation activities (Gully et al., 2002).

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Willingness to depend on others also facilitates knowledge sharing and integration within the team (Srivastava et al., 2006). When individuals are open to relying on their teammates' expertise and insights, they are more likely to share their own knowledge and engage in collaborative ideation actively (Edmondson et al., 2004). This exchange of ideas strengthens team-oriented commitment as members recognize the value of their collective knowledge (Srivastava et al., 2006). Moreover, dependence on others can enhance a team's collective efficacy, which refers to the shared belief in the team's ability to achieve its goals (Gully et al., 2002). When individuals are willing to rely on their teammates, they are more likely to believe in the team's capacity to generate creative ideas and solutions, boosting team-oriented commitment (Gully et al., 2002).

Finally, willingness to depend on others can lead to increased social support and emotional bonds within the team (Bishop and Scott, 2000). When individuals feel comfortable relying on their teammates, they are more likely to develop stronger interpersonal relationships and a sense of camaraderie (Bishop et al., 2000). These emotional connections can enhance team-oriented commitment, as members feel a sense of belonging and loyalty to the team (Bishop and Scott, 2000). Also, when individuals are willing to depend on others, it can reduce individualistic behaviour and promote a more collaborative mindset (Choi and Yoon, 2018). Accordingly, we hypothesize a positive relationship between the willingness to depend on others and team-oriented commitment.

> *H4:* Willingness to depend on others is positively associated with team-oriented commitment to collaborative ideation activities.

Team-Oriented Commitment and Motivation to Contribute

Team-oriented commitment refers to the emotional attachment and sense of identification that individuals feel towards their team (Meyer and Allen, 1991). When team members are strongly committed to their team, they experience a deep sense of belonging and alignment with the team's goals and values (Mowday et al., 1979). This shared sense of purpose can create a motivational force that drives individuals to actively participate and contribute to team activities, including collaborative ideation sessions (Bishop et al., 2000). Moreover, team-oriented commitment fosters a supportive and psychologically safe environment within the team (Edmondson, 1999). When individuals feel that their teammates are committed and invested in the team's success, they are more likely to feel comfortable sharing their ideas and opinions without fear of negative consequences (Lechner and Tobias Mortlock, 2022). This supportive atmosphere encourages open communication and active participation in ideation activities, as team members feel valued and supported by their colleagues (Perrmann-Graham et al., 2022).

Team-oriented commitment can also enhance individuals' belief in the team's collective ability to generate valuable ideas and solve problems (Gully et al., 2002). This sense of collective efficacy instils confidence in team members that their efforts will lead to successful outcomes (Bandura, 1977). When individuals believe that their contributions

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to collaborative ideation will make a meaningful impact, they are more likely to be motivated to actively participate and share their ideas (Tierney and Farmer, 2002). This increased motivation stems from a sense of reciprocity and social exchange, where individuals feel compelled to match the efforts and contributions of their committed teammates (Blau, 1986). Accordingly, we hypothesize:

> *H5:* Team-oriented commitment is positively associated with the motivation to contribute to a collaborative ideation activity.

AI Social Presence and Motivation to Contribute

A key challenge in virtual, text-based collaboration settings is the lack of physical presence and visual cues of teammates (Molinillo et al., 2018; Srivastava and Chandra, 2018). Low social presence reduces the immediate perception and awareness of team member actions (Haines, 2021) and affects collaboration in various ways. Lowered social presence also results in loss of motivation to contribute to the team (Pinsonneault et al., 1999; Shepherd et al., 1995), difficulty in organizing problems within the team (Piccoli and Ives, 2003), and failure to complete assigned tasks (Sarker and Sahay, 2003). In addition, low social presence also results in decreased satisfaction within the team (Malhotra and Majchrzak, 2012) and negatively affects team performance (Malhotra and Majchrzak, 2012).

While previous research has indicated a direct effect of social presence on motivation (Sung and Mayer, 2012; Wang et al., 2022), we extrapolate from TPB and SPT to propose that the influence of perceived social presence, particularly that of AI teammates, on team members' motivation is more nuanced and indirect. Specifically, we posit that the relationship between social presence and motivation to contribute is mediated by team-oriented commitment and willingness to depend on others. SPT provides a foundational framework to understand these dynamics (Garrison et al., 2001; Short et al., 1976). We argue that team members' social presence can foster a sense of trust and psychological safety (Edmondson, 1999; Lechner and Tobias Mortlock, 2022), leading to increased team-oriented commitment (Bishop and Scott, 2000). As individuals perceive AI as a supportive and reliable teammate, they are more likely to feel emotionally attached and dedicated to the team's goals and values (Meyer and Allen, 1991; Wang et al., 2022). Moreover, the social presence of AI can facilitate the development of interpersonal relationships and emotional bonds within virtual teams (Kreijns et al., 2003; Wang et al., 2022). When team members experience a strong AI presence, they are more likely to engage in social interactions, share personal experiences, and develop a sense of camaraderie with both human and AI teammates (Tsai et al., 2021). These emotional connections can enhance team cohesion and increase individuals' willingness to depend on their teammates, including AI, during online collaborative ideation activities (Garrison, 2011; Molinillo et al., 2021). As team members feel more connected and invested in one another, they are more likely to trust and rely on each other's ideas and contributions, regardless of whether they come from human or AI teammates (Kreijns et al., 2003).

TPB further supports the mediating role of team-oriented commitment and willingness to depend on others in the relationship between social presence of AI and motivation to contribute (Ajzen, 1991). Social presence of AI positively influences team members' attitudes

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towards both human and AI teammates, as they perceive AI as a real and trustworthy collaborator (Ng et al., 2020; Pitardi and Marriott, 2021). Social presence can contribute to the development of subjective norms that support interdependence and collaboration within the team, including reliance on AI (Kreijns et al., 2003). These positive attitudes and subjective norms, in turn, promote a sense of accountability and responsibility among team members in online collaborative ideation (Munnukka et al., 2022). In addition, such emotional investments make the team members more likely to engage in active and meaningful discourse, ask questions, and provide feedback to both human and AI teammates (Rourke et al., 1999). This can foster a shared understanding among team members and enhance their motivation to contribute their expertise and insights to the collaborative ideation process (Sung and Mayer, 2012). Thus, we hypothesize:

> *H6:* The relationship between social presence and motivation to contribute is fully mediated by team-oriented commitment and willingness to depend on others.

OVERVIEW OF STUDIES

We conducted three empirical studies to examine the relation between social presence and the motivation to contribute via willingness to depend on others and team-oriented commitment. We also explored whether participants perceive any difference in social presence when collaborating with an AI teammate (vs. with a human) and if these differences could be alleviated by increasing the familiarity and understandability of an AI teammate. In study 1, we tested H1 and H3–H6. In study 2, we tested H1–H6. Study 3 represented an alternate experimental setting, using psychophysiological measures obtained from analysing participants' facial expressions and skin responses to confirm H1 and H3–H6 and overcome the potential limitations of self-reported measures (Gasper and Clore, 2000). Table I provides an overview of our studies, and Figure 3 provides a visual overview of the order of the studies and of the overall process followed for all three studies.[5]

Table I. Overview of our studies

	Study 1	*Study 2*	*Study 3*
Hypothesis	H1; H3–H6	H1–H6	H1; H3–H6
Experimental design	Between-subject, online experiment	Between-subject, online experiment	Between-subject, lab experiment
Method of data collection	Questionnaire, usage data	Questionnaire, usage data	Galvanic skin response (GSR), facial expression data, questionnaire
Data analysis	*T*-test, PLS analysis	*T*-test, PLS analysis	*T*-test, PLS analysis
Sample	mTurk,[a] n = 100	mTurk, n = 150	Students, n = 81

[a]Amazon's Mechanical Turk.

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Experiment Procedure

All three studies were conducted on *Collaboration Space*, a web-based platform developed for this study, incorporating text-based collaboration features. Online experiments are especially useful to study human behaviour as they have high internal validity due to having full control (Fink, 2022). *Collaboration Space* simulated a team of five geographically dispersed team members (including the research participant) in a synchronous text-based brainstorming session. All team members, other than the participant, were simulations designed to mimic intelligent actors within the collaboration. Like typical generative AI systems, they generated text (i.e., messages) during the experimental procedure. All simulated participants (actors) had gender-neutral names: Taylor, Kim, Alex, and Sam. *Collaboration Space* was either part of an online survey environment (studies 1 and 2) or was shown directly to the participants in the behavioural lab (study 3). To avoid confounds, *Collaboration Space* utilized predetermined texts for all conditions instead of a generative AI-based text. *Collaboration Space* incorporated a four-step procedure (screenshots and further material of each step can be found in the supplementary material [section 1]):

1. *Introduction*: *Collaboration Space* showed a website with introductory text describing how participants could engage in brainstorming with other people with the assistance of a facilitator. Participants were asked to enter their names before they were allowed to proceed further.
2. *Team formation*: The second screen simulated team formation. Participants were given the impression of being grouped with other participants and a facilitator. The team formation took 1 minute to complete as actors joined one at a time.
3. *Participant introduction*: The third screen simulated the introduction of each participant. The facilitators introduced themselves to the participants and asked them to introduce

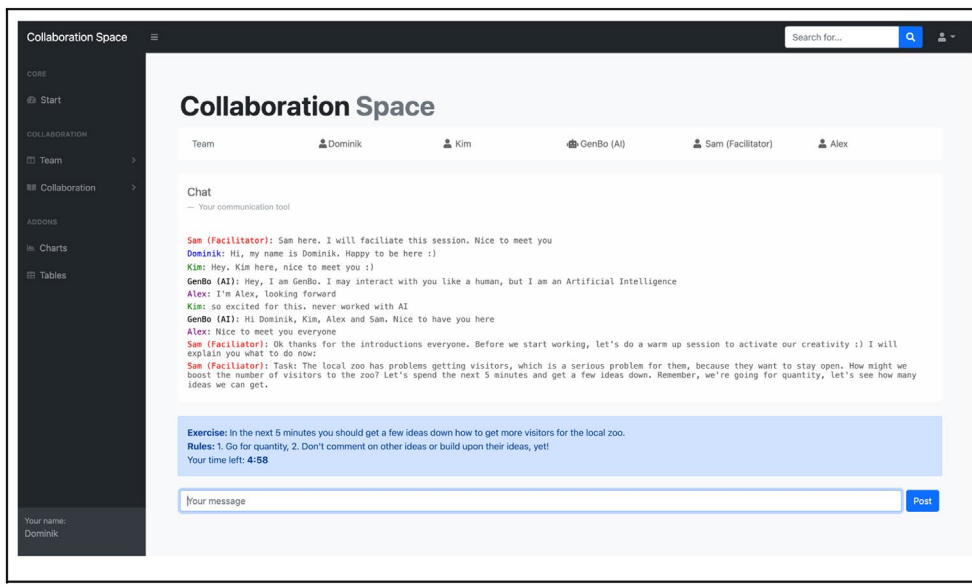

Figure 2. Screenshot of collaboration space (brainstorming step) [Colour figure can be viewed at wileyonlinelibrary.com]

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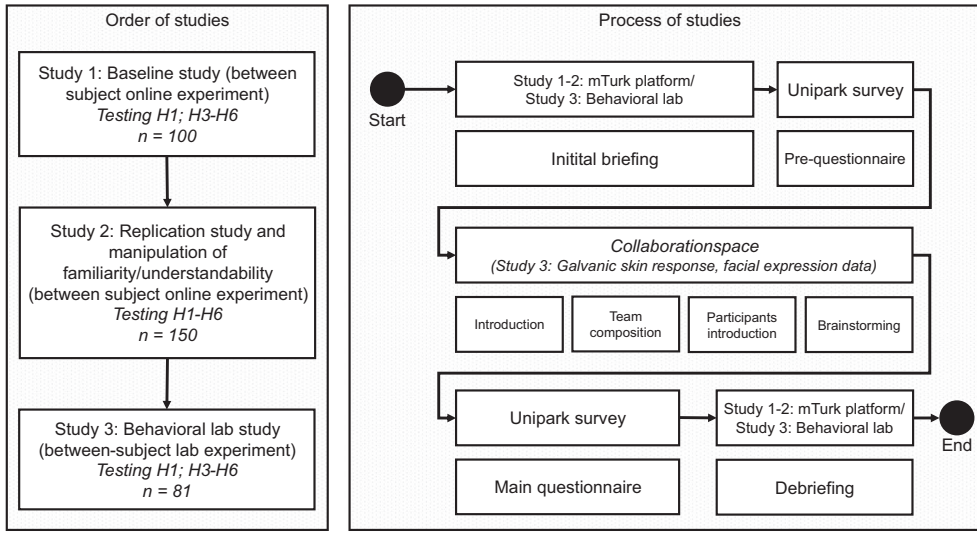

Figure 3. Overview of study order and process of the studies **$p < 0.01$; ***$p < 0.001$.

themselves. While the research participants introduced themselves, the actors' introductions were performed by the system to mimic the team members.

4. *Collaborative brainstorming*: On the final screen (Figure 2), the participants were asked to participate in a collaborative brainstorming task – *helping the local zoo attract more visitors*. The brainstorming process took place in a chat window, and all actors generated ideas in an analogous way as generative AI. For example, actor 'Kim' generated the idea: 'Another idea could be skywalks. Walking on bridges over them or between them'. To ensure experimental consistency, the actors were programmed to deliver uniform conversational quality across all sessions, reflecting a standardized mode of interaction. To mimic relevant and coherent dialogue, the actor's ideas were developed based on an actual recorded brainstorming session conducted and recorded by the company IDEO.[6]

The facilitator initiated the brainstorming activity by introducing the task and explaining the ground rules, notable one of which was to inform the participants that they were not allowed to build upon each other's ideas and had to generate new ideas. This protocol is informed by ideation literature, which underscores the value of generating a wide array of unique ideas during the early stages of brainstorming (i.e., Heslin, 2009; Paulus and Brown, 2007; VanGundy, 1984). Such an approach mirrors real-world scenarios where individual ideation precedes group synthesis, ensuring a diverse range of concepts before collaborative refinement (e.g., the 6-3-5 brainwriting method (Heslin, 2009)). Adhering to this rule allowed us to focus on the generation of unique ideas, a central aspect of brainstorming, while maintaining full control over the experimental conditions. We deliberately opted for this approach to highlight the individual contributions in a collaborative context, which is a crucial facet of early-stage brainstorming. This protocol was meant to emphasize the diversity of unique ideas contributed by each team member and to retain the standardization of

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conditions as much as possible. The collaborative brainstorming activity lasted for 5 minutes.[7]

STUDY 1

Objective

Study 1 was a between-subjects experiment to examine whether social presence is associated with motivation to contribute and whether this relationship can be explained through willingness to depend on others and team-oriented commitment. This study also examined whether people experience reduced social presence when interacting with an AI teammate. Participants were randomly assigned to either the experimental condition or the control condition.

Procedure

In study 1, participants from both control groups and experimental groups were informed that they would be brainstorming with different team members before being redirected to *Collaboration Space*. In addition, participants in the control group were informed that they would be collaborating with human team members. Conversely, in the experimental group, one team member revealed itself as GenBo, an AI bot. GenBo was part of the team formation process in the experimental group. In the control group, Taylor (a 'human' actor) was part of the formation process. This design aimed to isolate the effect of explicitly introducing an AI presence within the team, allowing us to examine its impact on perceived social presence and its association with participants' motivation to contribute. In the experimental group, GenBo introduced itself as an AI-based system ('Hey, I am GenBo. I may interact with you like a human, but I am an Artificial Intelligence Bot'.). In the control group, Taylor was introduced as a person ('Hey, I am Taylor. Nice to meet you guys. Looking forward to our brainstorming session'.). The execution of the brainstorming activity was identical for both the experiment and control conditions, that is, GenBo and Taylor contributed exactly the same ideas.

To address concerns about the script-based interaction, the teammates' responses were developed with a variety of potential replies, enabling them to respond to a range of potential inputs from participants. This was achieved through implementing a system that was designed to make the interaction as natural and collaboration-focused as possible. In addition, we conducted manipulation checks[8] post-experiment to ensure that participants were not doubting the identity of their teammates. These tests are commonly used to assess whether participants correctly infer the purpose of the study, particularly in experimental designs where manipulation awareness might bias responses (Kimmel, 2009; Orne, 1962).

Collaboration Space was embedded in a survey-based environment (Unipark software) in which participants were asked to complete a pre-survey prior to being redirected to *Collaboration Space*. They were also asked to complete a post-survey immediately after the brainstorming activity. Prior to entering the *Collaboration Space*, participants were also briefed

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that they were going to test for a new idea generation and collaboration website. They were told that the aim of the study was to test the functionality and design of the website. Subsequently, the participants were redirected to the survey on the Unipark software. Figure 3 shows the process of study 1, as well as an overview of all three studies.

Measures

The pre- and post-survey questionnaires each consisted of an adaptation of the five-item social presence scale by Qiu and Benbasat (2009), a four-item willingness to depend on others scale adapted from McKnight et al. (2002), a seven-item team-oriented commitment scale from Van den Heuvel et al. (1995), a five-item motivation to contribute scale (Lin, 2006), and demographic questions. All measures were on a 5-point Likert-type scale (see the complete questionnaire in the Supporting Information section 6).

Several measures were taken to ensure that participants were actively involved in the experiment. First, we ensured that participants remained in the open browser window of *Collaboration Space* for the entire duration of the experiment; whenever participants closed or switched the browser window, the experiment automatically terminated, excluding those participants from the study. We also included several attention-check questions in the questionnaires.[9] Furthermore, in the post-experiment survey, we asked participants to provide the name of the AI teammate or facilitator, depending on which condition they were in. Finally, we asked the participants whether they thought that GenBo was indeed an AI teammate (and vice versa for the control group) and whether the participants thought that the brainstorming session was a genuine session with real participants. Both questions were intended as criteria to exclude participants who perceived the experiment as a simulation.

Participants

One hundred and eighty participants signed up to our study from mTurk. We chose mTurk for our study because it offers an integrated compensation system and provides access to a large pool of participants. Furthermore, research has shown that mTurk is a feasible and cost-effective way to gather psychometrically reliable data (Buhrmester et al., 2011; Coppock, 2019; Peer et al., 2014). However, some authors have raised concerns about mTurk data quality (Aruguete et al., 2019) and potential overreliance on a subset of mTurk workers (Chandler et al., 2019; Robinson et al., 2019). To address quality data concerns, we followed the guidelines by Keith et al. (2017) and Peer et al. (2014) and required all participants to have a 'Master' qualification, as defined by mTurk.[10] All participants were employed at the time of the study, ensuring relevance to organizational collaboration contexts.[11] Furthermore, we ensured that our population of interest was sufficiently broad. Finally, we removed 80 participants from our analysis because they did not pass our stringent attention or manipulation checks.[12] This resulted in a total of 100 participants (42.0 percent females; average age: 41.6 years) for study 1 (54 participants in the experimental condition and 46 participants in the control condition). We performed power determination analyses using G*Power 3.1 (Faul et al., 2007), based on the criterion of 0.80 for acceptable

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statistical power (Giner-Sorolla et al., 2024). For the PLS-SEM analysis, modelled as a fixed-effects linear regression (R^2 deviation from zero), the target sample size required to detect a medium effect ($f^2 = 0.15$) at $\alpha = 0.05$ was 77. Our actual sample size of 100 exceeded this threshold, resulting in an achieved power of 0.90. For the independent samples *t*-test on perceived social presence, the estimated sample size needed to detect an effect of $d = 0.50$ at $\alpha = 0.05$ was 51 participants per group. Our final group sizes of 54 and 46 were slightly unbalanced, with the control group slightly below this target. Nevertheless, the achieved power remained at 0.80, indicating that the sample size was sufficient for detecting the expected effect.[13] Our sample size of 100 thus exceeded the minimum required for both analyses. Demographics of the participants can be found in the Supporting Information [section 5].

Results

Our data analysis consisted of three steps: the analysis of reliability, the examination of significant differences, and the examination of the underlying mechanisms and relationships using partial least squares structural equation modelling (PLS-SEM). We employed PLS-SEM for our data analysis, given its suitability for exploratory research that seeks to build theoretical insights. PLS-SEM is particularly adept at handling complex models and is robust to smaller sample sizes, making it ideal for examining the relationships proposed in our study[14] (Evermann and Rönkkö, 2023; Gefen and Straub, 2005). Furthermore, to assess mediation effects within our model, we utilized a bootstrapping approach with 5000 samples. This non-parametric resampling technique allowed us to estimate indirect effects without relying on normal distribution assumptions, providing more reliable confidence intervals for our analysis (Williams et al., 2009). The reliability of all scales met the minimum required Cronbach's alpha value of 0.700 (Cortina, 1993) (*social presence* = 0.941; *willingness to depend on others* = 0.897; *team-oriented commitment* = 0.796; *motivation to contribute* = 0.744) (Table II).

We used confirmatory factor analysis to test the measures and found that all items had significant (significance level of $p \leq 0.001$) positive factors ranging from 0.72 to 0.89. The measurement model demonstrated a good fit: $\chi^2 = 12.45$, df = 12, p = 0.014, CFI = 0.96, TLI = 0.95, RMSEA = 0.045, 90% CI [0.015, 0.078], SRMR = 0.048.

Table II. Descriptive statistics and correlations for study 1 (N = 100)

Variable	*M*	*SD*	*1*	*2*	*3*	*4*
Social presence	3.56	0.72	(0.941)	0.80***	0.45***	0.29**
Willingness to depend	3.48	0.69	0.80***	(0.897)	0.42**	0.24*
Team commitment	3.71	0.64	0.45***	0.42**	(0.796)	0.86***
Motivation to contribute	3.66	0.78	0.29**	0.24*	0.86***	(0.744)

Notes: Values on the diagonal are Cronbach's alpha.
*p < 0.05; **p < 0.01; ***p < 0.001.

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Furthermore, we calculated composite reliability (CR) and average variance extracted (AVE). CR values ranged from 0.760 to 0.945 (social presence = 0.945, willingness to depend = 0.905, team commitment = 0.810, motivation = 0.760), and AVE values ranged from 0.510 to 0.680 (social presence = 0.680, willingness to depend = 0.610, team commitment = 0.520, motivation = 0.510), all exceeding recommended thresholds and indicating reliable factors with adequate convergent validity (Urbach and Ahlemann, 2010).

Prior to performing a sequential mediation analysis, we tested whether social presence affects motivation to contribute without introducing any mediators using linear regression with social presence as the independent variable and motivation to contribute as the dependent variable.[15] The results show that social presence was significantly associated with motivation to contribute, $R^2 = 0.217$, $F(1, 98) = 27.10$, $p < 0.001$, $\beta = 0.47$, 95% CI [0.29, 0.64]. The effect size was large, $f^2 = 0.28$. Then, we performed a sequential mediation analysis using SmartPLS3 to test H3–H6 with *social presence* as the independent variable, *willingness to depend on others* and *team-oriented commitment* as mediating variables, and *motivation to contribute* as the dependent variable. We also included the control variables of *gender*, *age*, and *education*. However, they did not have significant effects on our outcome variable of *motivation to contribute*.

We employed several statistical measures to assess whether our factor analysis model adequately described the data. The chi-square statistic was calculated, as were various fit indices including RMSEA, CFI, TLI, and SRMR (Hooper et al., 2008; Hu and Bentler, 1999). The results of the chi-squared goodness-of-fit test were significant, $\chi^2 = 16.88$, df = 15, $p = 0.005$, indicating a statistically significant deviation from a perfect model fit. However, this result is not uncommon in models with larger sample sizes, where even minor misspecifications can lead to significant chi-squared values. Therefore, model fit is evaluated using multiple indices (Hu and Bentler, 1999). The TLI was 0.97, which exceeds the 0.95 threshold for good model fit. The CFI was 0.94, suggesting good model fit. The RMSEA index was 0.009, 90% CI = [0.00, 0.05], which is well below the 0.08 threshold, indicating excellent model fit. Finally, the SRMR was 0.07, which is below the 0.08 cut-off, implying that the model adequately fits the data (Hooper et al., 2008; Hu and Bentler, 1999).

There was a significant direct effect of team-oriented commitment on the motivation to contribute ($\beta = 0.86$, $p < 0.001$), showing support for H5, and a significant direct effect of willingness to depend on team-oriented commitment ($\beta = 0.42$, SE = 0.12, $p = 0.002$, 95% CI [0.19, 0.66]) supporting H4. Furthermore, we found a significant direct effect of social presence on willingness to depend on others ($\beta = 0.80$, SE = 0.08, $p < 0.001$, 95% CI [0.64, 0.96]) supporting H3. We found a significant direct effect of social presence on the motivation to contribute ($\beta = 0.29$ SE = 0.11, $p = 0.009$, 95% CI [0.07, 0.51]). Additionally, the significant indirect effect of social presence on the motivation to contribute ($\beta = 0.30$, SE = 0.14, $p = 0.031$, 95% CI [0.03, 0.57]), mediated by the willingness to depend on others and team-oriented commitment, was of comparable magnitude. These results indicate partial rather than full mediation, providing partial support for H6. Figure 4 shows the sequential mediation model, including the significant direct and indirect effects.

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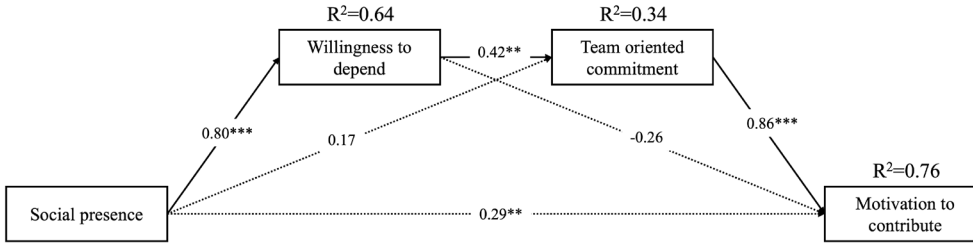

Figure 4. Sequential mediation model (Study 1) **p < 0.01; ***p < 0.001.

In the second step, we tested whether there was a significant difference between the two conditions regarding social presence. An independent samples *t*-test revealed significant differences between conditions in social presence measures (M_AI = 3.32, SD = 0.77; M_NoAI = 4.01, SD = 0.85), t(98) = 4.25, p < 0.001, d = 0.85, 95% CI [0.37, 1.01], representing a large effect size, supporting H1.

Discussion

The results of study 1 provide insight into the mechanisms underlying the effect of social presence on participants' motivation to contribute. We show that social presence was not as pronounced in the condition including an AI teammate. In addition, social presence was not directly associated with motivation to contribute. Instead, the association between social presence and motivation to contribute occurred indirectly via willingness to depend on others and team-oriented commitment. These results are in line with our hypotheses (H1 and H3–H6) as well as existing studies that show that social presence is an essential part of the text-based interaction with AI-based systems (Diederich et al., 2022; Elshan et al., 2022) which we further address in study 2.

While we observed a significant direct effect of social presence on motivation to contribute, our findings reveal that this relationship is more complex than a simple direct association. The presence of both direct (β = 0.29) and indirect effects (β = 0.30) of comparable magnitude suggests that social presence operates through multiple pathways to influence motivation. Importantly, the indirect pathway through willingness to depend on others and team-oriented commitment represents a critical mechanism that explains how social presence translates into collaborative behaviour, particularly in human–AI teams where traditional social dynamics may be disrupted.

STUDY 2

Objective

Building on study 1, study 2 further explored the dynamics of AI integration within text-based team collaboration. In study 2, we examined whether the increased social presence of an AI teammate is associated with an increased motivation to contribute through indirect relationships. Similar to study 1, study 2 was designed as a between-subjects

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experiment on *Collaboration Space* with two conditions. Participants were randomly assigned to the experimental group or the control group.

Procedure

As in study 1, in both the experimental and control groups, participants were told on the mTurk platform that they would collaborate with other team members on a brainstorming task on *Collaboration Space*. In the experimental group, participants were additionally informed that one of these team members would be GenBo, an AI bot. This early disclosure was designed to increase familiarity and understandability of the AI and test its effect on participants perceived social presence and the relationship between social presence and their subsequent motivation to contribute. As mentioned, familiarity arises from prior direct or indirect interaction and consequently can be increased by an earlier introduction of GenBo (Komiak and Benbasat, 2006). After its initial introduction on mTurk, GenBo was described in greater detail on *Collaboration Space* to increase its understandability as follows:

> *GenBo is an intelligent system that uses the newest technological advancements in the field of AI. Using natural language processing (NLP), GenBo can understand the task provided by the facilitator. Furthermore, GenBo has a huge database that it uses to combine existing information and form new ideas that make contributions toward the task. GenBo can express its ideas and 'thoughts' in written natural language.*

To ensure the effectiveness of our familiarity and understandability manipulations in study 2, we conducted pilot studies prior to the main experiment. Participants in these pilot studies confirmed that the manipulation effectively increased their perception of familiarity ($t(48) = 3.75$, $p < 0.001$) and understandability ($t(48) = 4.12$, $p < 0.001$) of the AI teammate compared to the baseline condition. Given the clear results from these pilot validations, we did not include additional manipulation checks in the main experiment, allowing us to maintain a streamlined focus on the core relationships under study.

In addition, during the introduction step on *Collaboration Space*, the participants were asked to introduce themselves in greater detail and focus on their strengths in terms of idea generation. In this step, GenBo's introduction was extended such that it explained its functionalities and ways of working in greater detail. Other than these changes, the procedure of study 2's procedure was identical to study 1 (see Figure 3). A detailed description of study 2 can be found in the Supporting Information (Section 1.2).

Participants

We recruited 172 participants from mTurk (study 1 participants were excluded from study 2). All participants had met the mTurk Master qualification. As in study 1, all participants were currently employed, providing a relevant sample for investigating workplace collaboration with AI teammates.[16] We removed 22 participants from our analysis,[17] which resulted in a dataset of 150 participants with 77 participants in the experimental group and 73 participants in the control group. We performed power determination

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analyses using G*Power 3.1 (Faul et al., 2007), again using 0.80 as the threshold for acceptable statistical power (Giner-Sorolla et al., 2024). For the PLS-SEM analysis, modelled as a fixed-effects linear regression (R^2 deviation from zero), the target sample size required to detect a medium effect ($f^2 = 0.15$) at $\alpha = 0.05$ was 77. With a total sample size of 150, the achieved power was 0.98, well above the recommended criterion. For the independent samples *t*-test comparing perceived social presence (in the experimental group) between study 1 (n = 54) and study 2 (n = 77), the estimated sample size needed to detect a medium effect (d = 0.50) at $\alpha = 0.05$ was 102 participants in total. The actual sample of 131 yielded a power of 0.88. For the independent samples *t*-test comparing perceived social presence between experimental and control conditions (n = 77 and n = 73), the same parameters applied, and the achieved power was 0.92. The average age of the participants was 38 years. Fifty-five participants were female, 93 were male, and two preferred not to disclose. The demographics of the participants are shown in the Supporting Information, Section 5.

Results

As with study 1, the data analysis consisted of three steps: an analysis of reliability, an examination of significant differences, and an examination of underlying mechanisms and relationships using PLS-SEM. The reliability of all scales met the minimum required Cronbach's alpha value of 0.700 (Cortina, 1993) (*social presence* = 0.838; *willingness to depend on others* = 0.814; *team-oriented commitment* = 0.705; *motivation to contribute* = 0.749) (Table III).

Confirmatory factor analysis revealed that all items had significant factor loadings exceeding 0.700 ($p \leq 0.001$), ranging from 0.71 to 0.87. The measurement model demonstrated acceptable fit: $\chi^2 = 18.50$, df = 16, p = 0.005, CFI = 0.92, TLI = 0.96, RMSEA = 0.020, 90% CI [0.001, 0.042], SRMR = 0.040. Composite reliability (CR) and average variance extracted (AVE) values further confirmed measurement quality. CR values ranged from 0.745 to 0.875 (social presence = 0.875, willingness to depend = 0.850, team commitment = 0.745, motivation = 0.785), and AVE values ranged from 0.500 to 0.580 (social presence = 0.580, willingness to depend = 0.540, team commitment = 0.500, motivation = 0.550), all meeting recommended thresholds for convergent validity.

Table III. Descriptive statistics and correlations for Study 2 (N = 150)

Variable	*M*	*SD*	*1*	*2*	*3*	*4*
Social presence	3.62	0.70	(0.838)	0.62***	0.36***	0.21**
Willingness to depend	3.54	0.66	0.62***	(0.814)	0.40***	0.22**
Team commitment	3.75	0.61	0.36***	0.40***	(0.705)	0.72***
Motivation to contribute	3.69	0.75	0.21**	0.22**	0.72***	(0.749)

Notes: Values on the diagonal are Cronbach's alpha.
*p < 0.05; **p < 0.01; ***p < 0.001.

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Following the same procedure as in study 1, we calculated chi-square, as well as the RMSEA, the CFI, the TLI, and the SRMR (Hooper et al., 2008). The results of the chi-square goodness-of-fit test were significant, $\chi^2 = 18.50$, $p = 0.005$, indicating a statistically significant deviation from a perfect model fit. However, this result is not uncommon in models with larger sample sizes, where even minor misspecifications can lead to significant chi-square values. Therefore, model fit is evaluated using multiple indices (Hu and Bentler, 1999). The TLI was 0.96, which exceeds the 0.95 threshold for good model fit. The CFI was 0.92, suggesting acceptable model fit. The RMSEA index was 0.02, which is well below the 0.08 threshold, indicating excellent model fit. Finally, the SRMR was 0.04, which is below the 0.05 cut-off, implying that the model adequately fits the data (Hooper et al., 2008; Hu and Bentler, 1999).

In the second step, we tested H2, which proposed that team members would experience higher social presence when working with a familiar and understandable AI teammate than with an unknown, unfamiliar one. In study 1, participants interacted with GenBo 'out of the box', receiving no background information or explanatory cues (unfamiliar AI). In study 2, we provided a tutorial and an explanation of GenBo's capabilities and limits (i.e., understandability). Also, GenBo was introduced earlier to the participants than in study 1 (i.e., familiarity). Social presence ratings were much higher for the familiar GenBo ($M = 4.01$) of study 2 compared to the unfamiliar GenBo ($M = 3.28$) of study 1, $p < 0.001$; this difference was medium to large, $d = 0.54$, 95% CI [0.28, 0.80], supporting H2. To examine the AI compared to a human teammate, we analysed social presence ratings for GenBo and the human teammate Taylor within study 2. Although Taylor still elicited a higher social presence, the gap between GenBo ($M_GenBo = 4.01$, $SD = 0.86$) and Taylor ($M_Taylor = 4.23$, $SD = 0.90$) was markedly smaller than in study 1 ($t(145) = 1.52$, $p = 0.131$, $d = 0.25$, 95% CI [0.07, 0.33]). This difference was statistically significant but small in magnitude, indicating that the familiarity and understandability intervention substantially improved perceptions of social presence for the AI teammate, reducing the gap to the human condition.

To confirm whether the same relationships observed in study 1 also emerged in study 2, we performed a sequential mediation analysis (using the same control variables as in study 1). The results confirm our previously constructed sequential mediation model: the direct effect of social presence on willingness to depend on others is significant ($\beta = 0.80$, $p < 0.001$, 95% CI [0.62, 0.98]), as is the direct effect of willingness to depend on others on team-oriented commitment ($\beta = 0.42$, $p = 0.003$, 95% CI [0.15, 0.69]), supporting H3 and H4. Furthermore, the direct effect of team-oriented commitment on motivation to contribute is significant ($\beta = 0.88$, $p < 0.001$, 95% CI [0.72, 1.04]), further supporting H5. The direct effect of social presence on motivation to contribute was $\beta = -0.14$, $SE = 0.09$, $p = 0.124$, 95% CI [−0.32, 0.04], indicating a non-significant direct relationship. The indirect effect of social presence on motivation to contribute is significant ($\beta = 0.30$, $p = 0.030$, 95% CI [0.05, 0.56]) and mediated by willingness to depend on others and team-oriented commitment, providing support for H6. Figure 5 shows the sequential mediation model including the significant direct and indirect paths of study 2.

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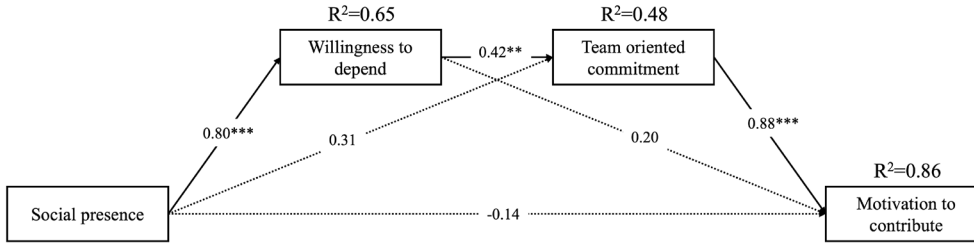

Figure 5. Sequential mediation analysis (Study 2) *p < 0.05.

Discussion

The results of study 2 suggest that the manipulation (i.e., the increase of familiarity and understandability of GenBo) is associated with increased social presence and with a substantial reduction in the difference in social presence between GenBo and Taylor. Furthermore, our findings reveal that perceived social presence can be enhanced by introducing the AI teammate early on and providing a clear, comprehensive explanation of how the AI operates. This stands in contrast to the commonly used strategy of anthropomorphizing AI agents to increase social presence (Seeger et al., 2021). Rather than relying on human-like characteristics, transparency in AI functionality appears to foster a more authentic and meaningful sense of presence, as participants were better able to understand and engage with the AI teammate (Mahmud et al., 2022; Van Hooijdonk, 2021). This finding contributes to the human–AI collaboration literature by suggesting that clear communication about AI capabilities may be more effective in promoting social presence than attempts to make AI appear human-like. Furthermore, an analysis of our sequential mediation model reveals the underlying mechanisms and suggests that there is only an indirect relationship between social presence and the motivation to contribute, supporting H6. Although this mediation model provided insightful information about the underlying relationships, certain limitations remained in the survey method such as the measurement of self-reported measures – that is, the perception of willingness to depend on others, team-oriented commitment, and the motivation to contribute. We addressed this limitation in our third study by conducting a lab experiment using psychophysiological measures to uncover specific measurable behavioural indicators for these constructs.

STUDY 3

Objectives

To overcome the potential limitations of self-reported measures (Gasper and Clore, 2000), we conducted an additional study with objective measures. In this study, in addition to self-reported measures, we measured the dependent and independent variables through psychophysiological proxies, specifically participants' facial expressions and galvanic skin response. Psychophysiological measurements, as proposed by

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Dimoka et al. (2012), offer a robust alternative to self-reported measures. This approach helps address potential biases inherent in perceptual data and ensures the reliability of our findings.

Procedure

The empirical context of study 3 was similar to studies 1 and 2. The experiment was conducted in the behavioural lab at a major European university and was supervised by two of the authors. Each participant was allocated a 30-minute time slot for the experiment. After setting the participants up with the apparatus to measure their physiological responses, the researchers explained the experiment in a manner consistent with the online experiments. The participants were told that they would join an online platform called *Collaboration Space* and collaborate with other preassigned online team members on a brainstorming task. Once the participants joined the space, the researchers moved to an adjacent room with a one-way window with a clear view of the participants' monitors. They also observed the participants via video during the duration of the experiment.

Measures

Emotional arousal has been a popular indicator of social presence in computer-mediated communication (Kojima et al., 2021). Research has shown that galvanic skin response (GSR – the level of measured conductivity provided by moisture on the skin's surface) indicates an increase in emotional arousal during interactions characterized by heightened social presence (Kojima et al., 2021; Riva et al., 2003). Furthermore, there is a distinct change in skin response when a person experiences emotional arousal (Bailey et al., 2009; Kojima et al., 2021; Petereit et al., 2022). Consequently, we used GSR to measure emotional arousal as a proxy for an increased sense of social presence. GSR was measured with a pair of silver chloride electrodes encapsulated in sodium chloride gel and placed on the participant's non-dominant hand. Changes in GSR are typically minor (greater than 0.05 microSiemens above baseline (Boucsein, 2012)). To account for the different amounts of time each participant was in the experiment, we calculated a baseline GSR by averaging the GSR readings taken while a researcher explained the study and got consent from the participants. To control pre-existing emotional states, we captured baseline GSR measurements before the collaborative task began. As a result, we calculated the baseline GSR and the change in GSR for each participant and condition. These baseline readings were subtracted from the experimental data to ensure that only task-related arousal was considered. Additionally, GSR was analysed in conjunction with self-reported social presence measures, allowing us to validate that the changes in GSR were linked to participants' perceptions of social presence rather than unrelated emotional states (e.g., excitement or stress). The controlled environment and consistent task structure further minimized potential confounding factors.

The relationship between the level of engagement in discussions and the degree of contributions that people make is mutual and reinforcing. While active engagement can lead to more valuable contributions (Bhatnagar, 2012; Tims et al., 2013), the act

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of making contributions can also deepen a person's engagement in the discussion. Therefore, engagement levels can be an effective proxy for people's motivation to contribute (Torrente et al., 2012). Research suggests a strong correlation between an individual's level of engagement in an activity and the expressions they display (Grafsgaard et al., 2013; Whitehill et al., 2014). When individuals are actively engaged, their faces often manifest various cues: focused attention may be reflected in furrowed brows or a steady gaze; enthusiasm might be displayed through wide eyes and a broad smile; conversely, confusion or frustration may result in a furrowed brow or a scowl. These expressions, collectively, serve as a 'psychophysiological mirror' that reflects the level and nature of a person's engagement. Given this correlation, we utilized facial expressions as a psychophysiological indicator of engagement and motivation in our study, capturing a more nuanced indication of engagement, extending beyond mere participation metrics to include the emotional and cognitive investment of participants in the activity at hand.

We analysed the recorded videos with the Affectiva Affdex (www.imotions.com) facial expression algorithm, which detects facial feature points and applies a set of rules based on psychological theories and statistical procedures to classify discrete emotional states and responses (Ekman and Friesen, 1978). The algorithm identifies a variety of points on the face and assesses how those points change as the person's face moves during the experiment, using distinct statistical procedures, facial databases, and facial feature points (for more information on the software, see iMotions 2016; for a validation study, see Stöckli et al. (2018)). This analysis technique has been used in a broad range of fields, such as marketing (Teixeira et al., 2014), entrepreneurship (Stroe et al., 2020), and psychology (Leppanen et al., 2017), but it is relatively new to IS research (Sipior, 2021).

We computed raw facial expression data from the videos with a threshold of 95, meaning we accepted only facial expressions that had at least a 95 percent chance of a human assessor rating the emotion the same as the Affdex algorithm. The threshold data are binary, with 1 indicating the probability of the participant expressing the emotion at 95 percent or higher and 0 indicating the probability of the participant expressing the emotion at less than 95 percent. A rating of 0 was interpreted as a lack of evidence of the emotion. The Affdex algorithm calculates the relative percentage of times that each emotion is displayed based on the binary threshold data by dividing the sum of those frames that scored 1 by the total number of frames. The data in this scoring procedure represent the percentage of times the participant displayed the specific facial expression. The sum of the percentage of times the participant displayed the tracked emotions was used to calculate affect.

Participants

We recruited 83 participants from a student pool. We performed power determination analyses using G*Power 3.1 (Faul et al., 2007), with 0.80 as the threshold for adequate statistical power (Giner-Sorolla et al., 2024). For the PLS-SEM analysis, modelled as a fixed-effects linear regression (R^2 deviation from zero), the target sample size required to detect a medium effect ($f^2 = 0.15$) at $\alpha = 0.05$ was 77. Our actual sample size of 83 exceeded this threshold,

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Table IV. Descriptive statistics and correlations for Study 3 (N = 81)

Variable	*M*	*SD*	*1*	*2*	*3*	*4*	*5*	*6*
Survey measures								
Social presence	3.66	0.71	(0.938)	0.80***	0.42**	0.29**	0.30	0.45***
Willingness to depend	3.58	0.68	0.80***	(0.884)	0.42**	0.24*	0.28*	0.38***
Team commitment	3.74	0.64	0.42**	0.42**	(0.781)	0.86***	0.26*	0.50***
Motivation to contribute	3.64	0.78	0.29**	0.24*	0.86***	(0.732)	0.27*	0.78***
Physiological measures								
Social presence (GSR)	3.03	0.68	0.30	0.28*	0.26*	0.27*	–	0.35**
Motivation to contribute (facial expressions)	3.84	0.92	0.45***	0.38***	0.50***	0.78***	0.35**	–

Notes: Values on the diagonal are Cronbach's alpha for survey measures. – indicates not applicable.
*$p < 0.05$; **$p < 0.01$; ***$p < 0.001$.

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yielding an achieved power of 0.84. For the dependent samples *t*-test comparing self-reported and physiological measures of motivation to contribute and social presence, the estimated sample size required to detect a medium effect (dz = 0.50) with $\alpha = 0.05$ (one-tailed) was 27. With 83 matched observations, the achieved power was 0.99. The total sample size of 83 thus exceeded the minimum required for all analyses. The participants were only required to have a good grasp of the English language. Two participants were removed from our analysis because they failed the manipulation checks. This resulted in a dataset of 81 (48.2 percent females; average age: 24.0 years) participants, with 41 in the experimental and 40 in the control condition. The demographics are shown in the Supporting Information [section 5].

Results

Our data analysis followed a similar pattern to the previous two studies: we began with an examination of reliability for the measures collected via the survey, moved to an examination of the significant differences in all measures (psychophysiological and survey data), and then concluded with an investigation of the underlying mechanisms and relationships using path analysis. All scale items had Cronbach's alpha values of greater than 0.700 (Cortina, 1993) (*social presence* = 0.88; *willingness to depend on others* = 0.83; *team-oriented commitment* = 0.71; *motivation to contribute* = 0.88) (Table IV).

The CFA supported the measurement model with all factor loadings exceeding 0.70 and reaching statistical significance (all $p < 0.001$). Composite reliability values ranged from 0.75 to 0.94, and AVE values ranged from 0.52 to 0.78, indicating adequate convergent validity (Urbach and Ahlemann, 2010). We employed several statistical measures to assess whether our factor analysis model adequately described the data. The chi-square statistic was calculated, as were various fit indices including RMSEA, CFI, TLI, and SRMR (Hooper et al., 2008; Hu and Bentler, 1999). The results of the chi-squared goodness-of-fit test were significant, $\chi^2 = 25.34$, $p = 0.010$, indicating a statistically significant deviation from a perfect model fit. However, this result is not uncommon in models with larger sample sizes, where even minor misspecifications can lead to significant chi-squared values. Therefore, model fit is evaluated using multiple indices (Hu and Bentler, 1999). The TLI was 0.95, which meets the threshold for good model fit (Hooper et al., 2008). The CFI was 0.93, suggesting acceptable model fit. The RMSEA index was 0.03, 90% CI = [0.00, 0.06], which is well below the 0.08 threshold, indicating good model fit. Finally, the SRMR was 0.03, which is below the 0.05 cut-off, implying that the model adequately fits the data (Hooper et al., 2008; Hu and Bentler, 1999).

We first analysed the data using survey measures to enable comparison with physiological measures. Using survey data, the direct effect of social presence on willingness to depend was $\beta = 0.80$, $p < 0.001$, 95% CI [0.5, 1.1], the effect of willingness to depend on team commitment was $\beta = 0.42$, $p = 0.002$, 95% CI [0.15, 0.69], and the effect of team commitment on motivation to contribute was $\beta = 0.86$, $p < 0.001$, 95% CI [0.55, 1.17]. The direct effect of social presence on motivation to contribute was $\beta = 0.29$, $p = 0.009$, 95% CI [0.07, 0.51]. The indirect effect through the mediators was $\beta = 0.30$, $p = 0.031$, 95% CI [0.03, 0.57]. Similar to study 1, both the direct and

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indirect effects of social presence on motivation to contribute are significant, partially supporting H6.

We then examined the model using psychophysiological measures. Supporting H5, team-oriented commitment was a strong predictor of the motivation to contribute ($\beta = 0.80$, $p = 0.040$, 95% CI [0.56, 1.04]). Supporting H4, willingness to depend on others was a predictor of team-oriented commitment ($\beta = 0.48$, $p = 0.045$, 95% CI [0.16, 0.68]). Furthermore, the direct effect of social presence on willingness to depend on others was positive ($\beta = 0.41$, $p = 0.050$, 95% CI [0.56, 1.04]), supporting H3. Lastly, the indirect effect of social presence on the motivation to contribute is significant ($\beta = 0.43$, $p = 0.031$, 95% CI [0.03, 0.57]), mediated by willingness to depend on and team-oriented commitment, supporting H6. The direct effect of social presence (measured by GSR) on motivation to contribute (measured by facial expressions) was $\beta = 0.07$, $SE = 0.15$, $p = 0.641$, 95% CI [−0.22, 0.36], indicating no significant direct relationship when using psychophysiological measures. Thus, the sequential mediation, that is, the indirect effect of social presence on motivation to contribute is supported by psychophysiological measures, supporting H6.

To validate our psychophysiological approach, we compared the survey data with the psychophysiological data. Self-reported motivation to contribute correlated strongly with the facial expression-based motivation measure ($r = 0.78$, $p < 0.001$). Self-reported social presence showed close alignment with the GSR-based social presence measure, with no significant difference between means ($M_{self} = 3.09$ vs. $M_{GSR} = 3.03$) and a small, nonsignificant association consistent with measurement equivalence ($r = 0.30$, $p > 0.05$). While GSR measures general emotional arousal, our study design controlled for potential confounds by capturing baseline GSR data and linking the physiological measures to self-reported social presence. These steps help to ensure that the observed changes in GSR are attributable to social presence rather than unrelated emotions, such as anxiety or excitement. Nevertheless, we acknowledge that GSR is a broad measure and may capture multiple emotional responses, which we address by combining it with subjective social presence assessments. We found a statistically significant difference between self-reported motivation to contribute ($M = 3.64$, $SD = 0.78$) and motivation measured by facial expression ($M = 3.84$, $SD = 0.92$). This difference, while statistically significant, represents a small effect size and likely reflects the complementary nature of these measurement approaches. Self-reported measures capture participants' conscious perceptions of their motivation, while facial expression analysis reflects emotional engagement that may occur below the threshold of conscious awareness. The consistency in directional relationships across both measurement types strengthens confidence in our findings, as each method provides unique but converging evidence for the underlying theoretical relationships. Figure 6 shows the sequential mediation model, including the significant direct and indirect paths of study 3.

Discussion

Our findings in study 3 show that psychophysiological measures confirm the findings of studies 1 and 2, providing further support for our hypotheses and research model.

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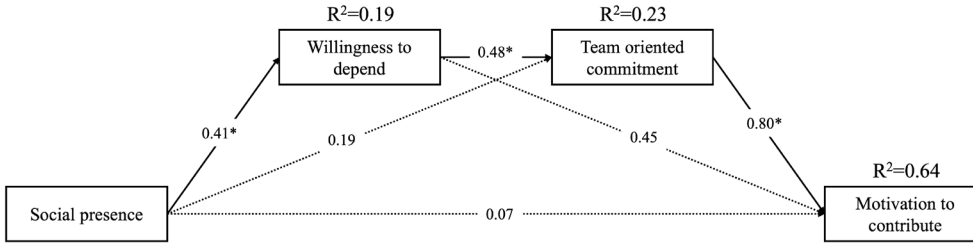

Figure 6. Sequential mediation analysis (Study 3)

Furthermore, while we found statistically significant differences between some measures, the overall pattern of relationships remained consistent across measurement types, supporting the robustness of our theoretical model. The consistency in directional relationships across self-reported and psychophysiological measures contributes to the validity and robustness of our findings, demonstrating that different operationalizations of our constructs yield consistent theoretical relationships.

GENERAL DISCUSSION

The results of our studies indicate that social presence is an essential element associated with the willingness to share ideas during online collaborative ideation, both in human and human–AI teams. The sequential mediation models of study 1 and study 2 reveal that motivation to contribute is associated with social presence, and this relationship is mediated by willingness to depend on others and team-oriented commitment. Study 3 further confirms our findings using psychophysiological measures (GSR and facial expressions). Providing further substantiation to the importance of social presence in computer-mediated communication (Rourke et al., 1999; Tu and McIsaac, 2002), our results deepen our understanding of the mechanisms related to social presence and underscore the importance of social presence in virtual collaboration.

Our results consistently show that social presence is both directly and indirectly associated with motivation to contribute (which is essential for effective brainstorming outcomes). While a direct relationship exists, our findings reveal that the indirect pathway through willingness to depend on others and team-oriented commitment represents an equally important mechanism, particularly in the context of human–AI collaboration where traditional assumptions about social presence may not fully apply. Furthermore, each study provides evidence of a direct association between social presence and willingness to depend on others, as well as a strong association between willingness to depend on others and team-oriented commitment. That is, a greater willingness of team members to rely on others is related to higher team-oriented commitment, as commitment is important for forming a coherent team with reliable members (Sheng et al., 2010). This added commitment to the team is associated with greater motivation to contribute, which in turn is linked to improved overall team performance (Chang et al., 2010; Sheng et al., 2010).

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In study 2, we manipulated the familiarity and understandability of the AI teammate and found social presence to be a critical independent variable. We thus show that an increase in social presence could result from the early introduction and detailed explanation of the AI teammate rather than from behavioural manipulation or through anthropomorphizing the AI teammate, as is commonly practised (Araujo, 2018; Lee et al., 2021; Qiu and Benbasat, 2009).

Theoretical Implications

Our findings advance the theoretical understanding of human–AI collaboration, drawing on the results of all three studies to demonstrate how AI teammates change collaborative dynamics in ways that go beyond the predictions of traditional human–human collaboration frameworks. Across our studies, we show that AI teammates can assume a role as proactive, autonomous agents, which fundamentally alters how collaboration unfolds. In contrast to conversational agents or reactive AI systems, the AI in our experiments was not just a passive tool responding to human prompts; rather, it actively contributed to the collaborative process. This contribution shifts the understanding of human–AI collaboration by revealing that AI teammates can shape team processes depending on how they are introduced and understood within the team. Our findings suggest that successful human–AI collaboration is not contingent on anthropomorphizing AI or making it appear human-like, as has often been theorized (Araujo, 2018; Dubosc et al., 2021; Konya-Baumbach et al., 2023; Munnukka et al., 2022; Seeger et al., 2021). Instead, fostering collaboration is about making the AI's role and capabilities transparent and understandable, allowing human teammates to adjust their behaviour and expectations accordingly. This aligns with a relational view of expertise, where productive collaboration with AI systems is not primarily achieved by emulating human cognition but by embedding such systems into meaningful and situated interactional contexts (Pakarinen and Huising, 2025). This also shifts the focus in human–AI collaboration research from simulating human attributes to optimizing AI transparency and functional integration. This reframing contributes to a broader theoretical discourse on how AI systems are integrated into teams and offers new directions for future research on designing AI systems that optimize team performance and social dynamics.

Furthermore, our results contribute to SPT as they expand current understanding of perceived social presence with respect to AI-based systems. When people collaborate, they often develop or maintain a personal connection through interpersonal exchanges, which is associated with an increased sense of social presence and a greater willingness to depend on each other (Weisberg et al., 2011). Building on earlier studies involving conversational agents (Araujo, 2018; Elshan et al., 2022; Lee et al., 2021; Qiu and Benbasat, 2009), we show that participants experience lower social presence towards AI teammates that generate ideas (Molinillo et al., 2018). Our results are consistent with previous findings on the role of social presence in collaborative efforts and extend the application of these findings to human–AI collaboration contexts (Bao et al., 2021; Dennis et al., 2023; Sharma et al., 2023; Wei et al., 2017). Thereby, we extend SPT by demonstrating its relevance for collaborative processes involving

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human and AI teammates. SPT is relevant not only in interaction scenarios such as communication between humans and conversational agents (Elshan et al., 2022) but also with respect to collaborative ideation with AI-based systems. Our results indicate that simply describing or naming an actor in the collaboration process as 'AI' is sufficient to induce less social presence, even if human-like features are identical. Thus, we further extend the ongoing discussion on anthropomorphism and human-like design features in AI systems (Elshan et al., 2022; Feine et al., 2019; Seeger et al., 2021), noting that focusing solely on these aspects does not fully address the complexities of fostering social presence within teams. While Seeger et al. (2021) advocate the use of a combination of verbal, nonverbal, and identity cues, our findings highlight the importance of enhancing understandability and familiarity alongside the anthropomorphic designs. This holistic approach is essential for effective human–AI collaboration, as it not only enhances the social dynamics within teams but also contributes to the overall functioning and performance of the team. In addition, this perspective broadens the scope of SPT by suggesting that the path to effective human–AI collaboration lies not only in emulating human-like characteristics but also in fostering a deep understanding and familiarity with AI capabilities and roles. In line with this, effective human–AI collaboration is rooted in interactional fluency and practical familiarity rather than static system attributes – a perspective emphasized by relational approaches to expertise (Pakarinen and Huising, 2025).

Our finding of both direct and indirect effects of social presence on motivation to contribute enriches rather than contradicts our theoretical framework. The presence of a direct effect alongside the mediated pathway suggests that social presence operates through multiple mechanisms in human–AI teams. While traditional SPT would predict a straightforward direct relationship, our results show that in human–AI collaboration contexts, the indirect pathway through team-oriented commitment and willingness to depend on others is equally important. This dual-pathway finding extends SPT by demonstrating that the mechanisms linking social presence to collaborative outcomes are more complex in hybrid human–AI teams than in traditional human-only teams.

Our findings further contribute to a deeper understanding of behavioural intentions and the relationship between beliefs and behaviour. By demonstrating the association between social presence and team members' behavioural intentions, we establish a connection between SPT and TPB within collaborative ideation. This linkage not only reaffirms but also expands the constructs established by Ajzen (1991), while adding depth to their application in team dynamics as examined by Wang et al. (2015) and Park and Yang (2012), providing a more comprehensive framework for examining how social dynamics are associated with individual commitments and actions within human–AI teams. By integrating the perceived social presence of AI as a factor influencing individuals' attitudes and subjective norms, our research enhances our understanding of TPB, particularly in the context of AI-enhanced teamwork. This addresses the call for a deeper exploration of the antecedents of collaborative behaviour in technology-mediated settings (Al-Lozi and Papazafeiropoulou, 2012). We also enrich the understanding of TPB in collaboration research by demonstrating the relationship between behavioural

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intentions and team phenomena, such as the willingness to depend on a team member and team-oriented commitment.

Furthermore, our findings complement and extend the findings of Dennis et al. (2023), Glover and Kim (2021) and Randrup et al. (2018) on the behavioural foundations of effective collaboration by illustrating how the introduction of AI teammates and their social presence is associated with changes in traditional team dynamics theories. We thus contribute to the evolving theoretical understanding of human–AI collaboration in team environments and its implications for insights on behaviours of individuals in teams. For instance, Baird et al. (2021) propose reconsidering the primacy of human agency in the context of agentic IS artefacts, which aligns with our findings on the evolving dynamics of human–AI collaboration. Their emphasis on delegation as a lens for understanding the relationship between human and agentic IS artefacts enriches our discussion of social presence. It highlights the evolving role of AIs from passive tools to active participants in collaborative efforts that convey social presence. This also aligns with emerging perspectives that emphasize how integrating AI into professional teams challenges established collaboration norms and requires new ways of legitimizing AI contributions within ongoing work practices (Faulconbridge et al., 2025).

Moreover, our studies highlight how the motivation to contribute is related to personal attitudes and beliefs towards the team. This extends the discussions initiated by Bunderson and Sutcliffe (2002) and Chae et al. (2015) into the domain of AI-augmented teams. This offers a novel perspective on the relevance of TPB, highlighting the complexity between technological integration, such as AI, into team behavioural dynamics. Finally, we also demonstrate that SPT and TPB are complementary by empirically illustrating the interplay between social presence, familiarity with AI team members, team beliefs, and behavioural intentions. This holistic approach not only addresses gaps identified by previous research but also establishes future research directions in collaboration technologies and team dynamics involving AI teammates.

Managerial Implications

Our findings have meaningful implications for organizations adopting AI-based systems, like generative AI, in collaboration scenarios such as idea generation. For instance, our results suggest that organizations should carefully design systems and work procedures to support social presence and prepare employees for AI teammates so that they can become familiar with the systems and understand how they work. This is in line with the current literature highlighting the importance of thoughtfully integrating AI into team dynamics (Seeber et al., 2019, 2020; Glikson and Woolley, 2020).

Our study advocates an approach that goes beyond anthropomorphic design and behaviour (Seeger et al., 2021) and highlights the need to foster an environment in which AI roles and capabilities are clear, understandable, and familiar to all team members (Harrison et al., 2003). Organizations should prioritize improving human factors such as trust and psychological safety while clearly articulating and conveying AI's role within

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teams. Such articulation is not only a design concern but also an ongoing organizational negotiation, as teams recalibrate how AI actors are perceived, legitimized, and integrated into collaborative routines (Faulconbridge et al., 2025). They should also focus on developing initiatives that increase employee willingness to rely on AI and foster team-oriented commitment, such as team-building activities that involve AI in collaborative tasks, demonstrating its reliability and usefulness in achieving team goals. A noteworthy observation from our research is that sole reliance on anthropomorphism to increase social presence in the individual design of AI teammates may not be appropriate in all collaboration contexts and not always be technically possible in text-based collaboration platforms. Despite the increasing technological sophistication, it is important to retain our focus on the human aspects that play a role in creating a social presence in online teams.

Echoing findings on augmenting human work with intelligent systems (Anthony et al., 2023; Berente et al., 2021), our research shows that promoting familiarity and understandability subsequently supports human–AI collaboration by influencing the motivation to contribute. This highlights the need for education and training programmes to ensure that team members are adequately prepared for and comfortable with AI collaboration (Hevner and Storey, 2022). Organizations should consider using AI systems to facilitate internal team communication, further integrating AI into daily workflows and increasing familiarity. This could include AI-assisted scheduling, task management, or even facilitating brainstorming sessions, gradually building trust and confidence.

Furthermore, our research demonstrates that simply naming an actor as an AI in virtual collaboration scenarios significantly affects co-creation, underscoring the importance of how AI is introduced. This aligns with the literature on the evolving role of AI in professional environments, where clear and effective communication of AI capabilities and roles can significantly improve collaborative outcomes (Anthony et al., 2023; Dennis et al., 2023). Moreover, our results show that the social presence of all actors in virtual collaborations can be improved not only by using anthropomorphic designs but also by increasing the understandability and familiarity with the AI team member.

Finally, our research highlights the critical need for organizations to enhance text-based collaboration platforms, such as Slack and Microsoft Teams, with AI technologies that prioritize familiarity and understandability. This is not just a short-term requirement but a long-term strategy for ensuring sustained success in human–AI collaboration. As AI systems become increasingly embedded into workflows, organizations should focus on scaling their integration by continuously updating platforms and work processes to keep pace with advancements in AI. Additionally, organizations must invest in long-term training and change management initiatives to ensure that employees remain adaptable and proficient in collaborating with evolving AI technologies. This approach will enable organizations to leverage AI's full potential, fostering an environment where both human and AI teammates can co-create effectively over time, particularly as remote and hybrid work become permanent fixtures in the workplace.

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LIMITATIONS AND FUTURE RESEARCH

When interpreting the findings of our research, a number of limitations should be considered, as they provide interesting opportunities for future research. As explained by Fink (2022), online experiments are artificial in nature due to the need to exercise high levels of control. Consequently, they have high internal validity but typically lower external validity. We addressed this drawback by developing and providing a realistic web platform and executing a series of manipulation checks. Nevertheless, our research setting may still have been perceived as artificial by the participants. Future studies could extend this concept through field studies with organizations or by employing more forms of AI-based systems.

Another limitation of our studies is that all variables were measured at the same point in time. While the experimental design provides some confidence in the relationships observed, the concurrent measurement of variables does not allow for definitive conclusions about causality. Future research could address this limitation by employing longitudinal or multi-wave designs to better capture the temporal dynamics of social presence and its effects on team dynamics. Additionally, experimental designs incorporating time-lagged measures could provide further evidence for the sequential relationships proposed in our model.

Furthermore, our samples in studies 1 and 2 were limited to mTurk participants, which may not be representative of the general population. To mitigate this, we ensured diversity within our mTurk samples through random sampling techniques and will aim for a more stratified approach in future work. The same concern is relevant to the students that participated in study 3: they predominantly represented university students from a single institution, which may have introduced selection and demographic biases, limiting the generalizability of our results to a broader population. This homogeneous sample in terms of age, educational background, and potentially socio-economic status may not fully reflect the dynamics within a more varied workforce. Future studies could examine the effects of AI team members on actual organizational employees. Such studies should aim to recruit participants from varied professional backgrounds and different types of organizations to ascertain the replicability of our findings in diverse settings.

Additionally, this study employed a specific task in a specific context (collaborative, virtual ideation for a zoo). Therefore, it remains to be determined whether the results of this study are transferable to other tasks and contexts. Further studies on human–AI collaboration in different contexts and tasks (e.g., solving puzzles, making decisions) should be conducted to determine whether similar associations between social presence and the motivation to contribute are observed.

Finally, the mode of communication in this study was exclusively via a chat interface. Future research could explore other interaction modalities (e.g., video chat, digital whiteboards). While we identified specific mechanisms that are associated with team members' motivation to contribute, we acknowledge other potential moderating factors, like AI anxiety or attitudes towards virtual ideation, which warrant further exploration.

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Our research examines how social presence is related to collaboration in hybrid human–AI teams through three empirical studies based on SPT and TPB. The results suggest that the relationship between social presence and motivation to contribute is not direct, but is associated with human factors such as team-oriented commitment and willingness to depend on others. These findings offer insights into relational mechanisms associated with effective collaboration in human–AI teams. They support the view that collaboration with AI evolves through situated interaction and shared practical understandings rather than merely through functional or anthropomorphic system attributes (Pakarinen and Huising, 2025). This contributes to the understanding of social presence in hybrid human–AI teams and also highlights the importance of familiarity and understandability of AI, while challenging the assumption that the reliance on anthropomorphic features in collaborative AI design is always beneficial. Our findings also speak to broader reconfigurations of collaborative labour in the age of AI. As Sarala et al. (2025) highlight, understanding how AI is integrated into everyday professional practices requires a shift in focus towards interactional and behavioural micro-mechanisms. By identifying how social presence shapes willingness to depend on AI teammates and commit to team goals, our study responds to this call for a grounded, micro-level understanding of hybrid teamwork.

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CONFLICT OF INTEREST STATEMENT

None.

NOTES

[1] https://slack.com/intl/en-in/features/ai.
[2] https://blogs.microsoft.com/blog/2023/03/16/introducing-microsoft-365-copilot-your-copilot-for-work/.
[3] https://support.microsoft.com/en-us/windows/welcome-to-copilot-in-windows-675708af-8c16-4675-afeb-85a5a476ccb0.
[4] https://www.microsoft.com/en-us/microsoft-365/blog/2024/09/16/microsoft-365-copilot-wave-2-pages-python-in-excel-and-agents/.
[5] Study 3 had a slightly different process as participants were physically present in the lab.
[6] We transcribed the brainstorming session conducted and recorded by IDEO and published on YouTube (https://www.youtube.com/watch?v=VvdJzeO9yN8). We developed the input of our AI actors based on the transcript with minor adjustments.
[7] The duration of the brainstorming session was intentionally limited to 5 minutes, a decision rooted in both empirical evidence (Haddou et al., 2014) and practical considerations. Short, time-constrained sessions have been shown to stimulate a high volume of idea generation (Osborn, 1953), a phenomenon

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also leveraged in the structured '6-3-5 brainwriting' method (VanGundy, 1984). Furthermore, the constraint of time is reflective of real-world organizational settings, where quick ideation is often necessary and ideas are produced more efficiently (Baer and Oldham, 2006; Schmitt et al., 2015). This design choice is substantiated by our results, indicating that the primary effects of interest manifest within this timeframe, in line with findings from Nijstad and De Dreu (2002) and Baer and Oldham (2006) on creativity under time pressure.

[8] To ensure the validity of our study's results, manipulation checks were implemented following the completion of the tasks. Participants were debriefed with a series of questions crafted to detect if they had perceived the true nature of the experiment. This debriefing aimed to ascertain that the participants' behaviour during the study was not influenced by any accurate guesses regarding the experimental conditions. Details of the debriefing process are further outlined in the Supporting Information.

[9] To assess participant engagement and ensure the integrity of our data, attention checks were strategically placed throughout the survey. These checks took the form of questions that required specific responses to confirm that participants were reading the instructions and questions carefully. Failure to answer these control questions correctly was used as an indicator of inattention. The questions and criteria for participant inclusion based on their responses are detailed in the Supporting Information (section 6).

[10] Information about Amazon mTurk master qualification: https://www.mturk.com/worker/help.

[11] Additional demographic characteristics are provided in the Supporting Information.

[12] 11 failed the basic attention check, 31 failed to provide the name of GenBo, 17 failed to provide the facilitator's name, and 21 were suspicious of the brainstorming simulation (as indicated by manipulation check).

[13] Detailed information on the justification for sample adequacy (power determination analysis) can be found in the Supporting Information (section 4).

[14] Further justification for the usage of PLS-SEM can be found in the Supporting Information (section 2).

[15] Simple linear regressions were repeated for study 2 and 3 and can be found in the Supporting Information, section 3.

[16] Full demographic details are reported in the Supporting Information.

[17] Three did not consent to our study, one failed the basic attention check and 18 were suspicious that the brainstorming was simulated (as indicated by manipulation check).

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